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1 CLF-CBF-QP Based Motion Planning Controller

This work implements a control strategy for safe trajectory generation in a 2D environment using a Control Lyapunov Function (CLF), Control Barrier Functions (CBFs), and a Quadratic Program (QP). The robot is modeled as a point-mass and operates in a field with Gaussian-shaped obstacle zones.

1.1 Control Lyapunov Function (CLF)

The Control Lyapunov Function ensures convergence to the goal:

$$V(x) = \frac{1}{2} \|x - x_g\|^2 \quad (1)$$

where $x \in \mathbb{R}^2$ is the robot position and x_g is the goal.

The Lie derivative along the control direction is:

$$L_f V = \nabla V^\top = (x - x_g)^\top \quad (2)$$

The corresponding CLF constraint is enforced in the QP:

$$L_f V \cdot u \leq -cV \quad (3)$$

where u is the control input and $c > 0$ is the CLF gain.

1.2 Control Barrier Functions (CBF)

For each obstacle centered at x_o , the barrier function is defined as a logarithmic function to enforce safety:

$$h(x) = \log(\|x - x_o\| - r_{\text{strict}}) \quad (4)$$

where r_{strict} is the red-zone radius (strict exclusion region).

The directional Lie derivative of h is shaped using a blend of gradient and tangent components to avoid oscillations near obstacles:

$$L_f h = \frac{\nabla h^\top}{\|x - x_o\| - r_{\text{strict}}} \cdot \text{scale} \quad (5)$$

The barrier constraint becomes:

$$-L_f h(x) \cdot u \leq \gamma h(x) \quad (6)$$

with $\gamma > 0$ as the barrier gain, which may adapt near the goal.

1.3 Quadratic Program (QP) Formulation

At each time step, the robot solves the following optimization problem:

$$\min_{u \in \mathbb{R}^2} \|u - u_{\text{ref}}\|^2 \quad (7)$$

$$\text{s.t. } L_f V \cdot u \leq -cV \quad (8)$$

$$-L_f h_1(x) \cdot u \leq \gamma h_1(x) \quad (9)$$

$$-L_f h_2(x) \cdot u \leq \gamma h_2(x) \quad (10)$$

where $u_{\text{ref}} = -1.0(x - x_g)$ is the nominal controller.

1.4 Velocity Damping via Exponential Decay

To enable smooth and safe traversal near obstacles, an exponential decay function modulates the maximum velocity based on proximity:

$$d = \min(\|x - x_{o_1}\|, \|x - x_{o_2}\|) \quad (11)$$

$$s = \frac{d - r_{\text{strict}}}{r_{\text{outer}} - r_{\text{strict}}}, \quad s \in [0, 1] \quad (12)$$

$$v_{\text{max}} = 1.5(1 - e^{-4s}) + 0.05 \quad (13)$$

This guarantees:

- High velocity when far from obstacles
- Smooth deceleration as the robot approaches obstacle centers
- A floor speed of 0.05 m/s in extreme proximity

1.5 Recovery Behavior for Stuck Conditions

If the robot slows excessively or turns away from the goal, recovery logic overrides the control to prevent stagnation:

$$u = 0.1 \cdot \frac{x_g - x}{\|x_g - x\|} \quad (14)$$

This ensures continued motion toward the goal.

1.6 Diagnostics and Metrics

The system logs and plots the following over time:

- **Velocity:** $\|u\|$ vs. time
- **Obstacle proximity:** $d = \min(\|x - x_{o_i}\|)$
- **Curvature:** turning rate computed as

$$\kappa(t) = \frac{\cos^{-1} \left(\frac{u_k^\top u_{k-1}}{\|u_k\| \cdot \|u_{k-1}\|} \right)}{\Delta t} \quad (15)$$

1.7 Conclusion

This CLF-CBF-QP framework achieves safe, smooth, and goal-directed navigation even in the presence of soft probabilistic obstacles. The exponential damping and directional barrier shaping allow the robot to maintain performance and avoid getting stuck in edge cases near obstacle centers.